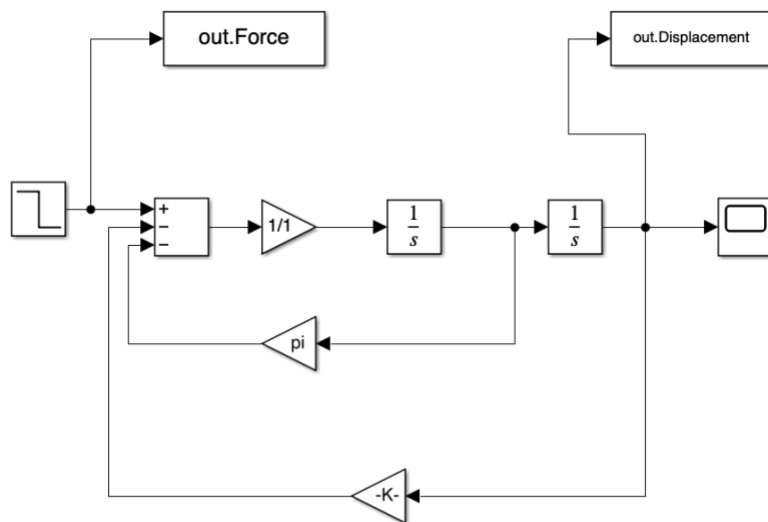


Lab9

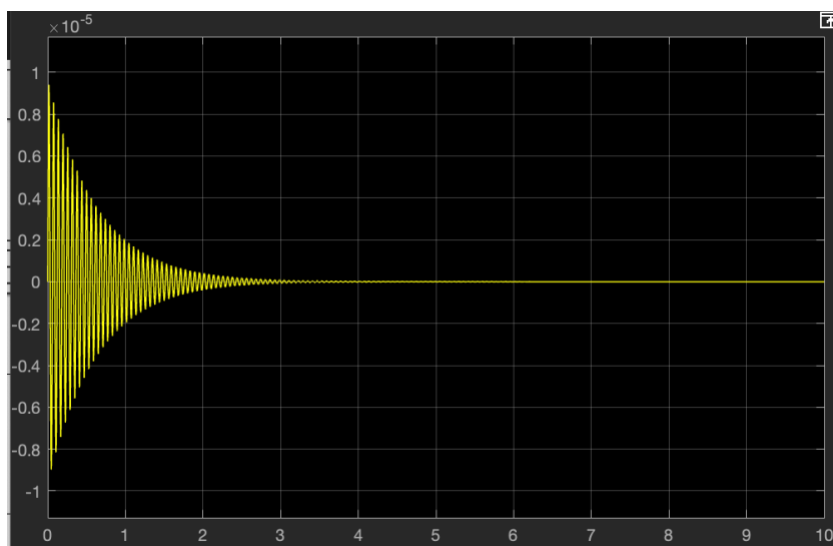
Simulate a single degree of freedom system in Simulink, export the vectors x , $\dot{x}$, $\ddot{x}$ and F , and then use the above equations to find the values of M , C and K



$$M = 1$$

$$C = \pi$$

$$K = 1089\pi^2$$



For the considered system with one degree of freedom, calculate the values of the coefficients a_1 and a_2

Denominator:

$$\text{den} = M + CT + KT^2 = 2 + 10 \cdot 0.001 + 100 \cdot (0.001)^2 \quad \text{den} = 2 + 0.01 + 0.0001 = 2.0101$$

$$\text{den} = M + CT + KT^2 = 2 + 10 \cdot 0.001 + 100 \cdot (0.001)^2 \quad \text{den} = 2 + 0.01 + 0.0001 = 2.0101$$

Numerators:

$$\text{num}_1 = 2M + CT = 4 + 0.01 = 4.01$$

$$\text{num}_2 = -M = -2 \quad \text{num}_1 = 2M + CT = 4 + 0.01 = 4.01$$

$$\text{num}_2 = -M = -2$$

Final values:

$$a_1 = \frac{4.01}{2.0101} \approx 1.99498$$

$$a_2 = \frac{-2}{2.0101} \approx -0.995$$

$$a_1 = \frac{4.01}{2.0101} \approx 1.99498 \quad a_2 = \frac{-2}{2.0101} \approx -0.995$$

Define the least squares problem for the above equation in a manner analogous to that in Problem 1). Find the parameter values a_1 and a_2 . In order to ensure correct results for tasks 2) and 3) change the solver to ODE1 and change the system parameters so that the system is stable
(example parameters $M=2$, $C=10$, $K=100$)

```
x = out.Displacement.Data; % timeseries → numeric array
```

```
N = length(x);
```

```
X_matrix = [x(2:N-1), x(1:N-2)];
```

```
Y_vector = x(3:N);
```

```
% Least Squares :
```

```
a = X_matrix \ Y_vector;
```

```
a1 = a(1);
```

```
a2 = a(2);
```

```
fprintf("a1 = %.5f\n", a1);
```

```
fprintf("a2 = %.5f\n", a2);
```

```
>> lab9mu
```

```
a1 = 1.98771
```

```
a2 = -0.99842
```

Use a second-order auto-regression (AR) model to find the difference equation for the system under analysis. Use Matlab's "iddata" function to create a data object and use it in conjunction with the "ar" function. Select the calculation method as the least squares method "ls". Compare the results with the results obtained from tasks 2) and 3). Due to the different definitions of the resulting formulas, the signs of the obtained coefficients will be different from those in the previous tasks, see the help of the ar function

```
>> simumus
```

Time domain damping (low): 0.9137

Time domain damping (high): 0.3624

Frequency domain damping (half-power) 1: 3.7000

Frequency domain damping (half-power) 2: 0.4348

iddata requires System Identification Toolbox.

Error in simumus (line 130)

```
DAT = iddata(res2, [], 1/fs);
```

